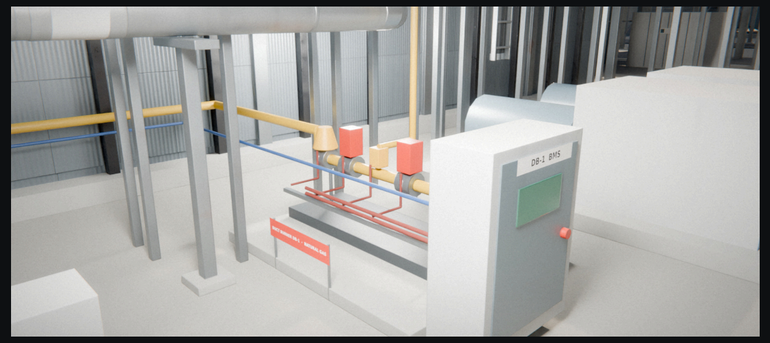
BOILERS: GREENFIELD & BROWNFIELD

One cable partner from the new HRSG to the package boiler to the outage that replaces a module: the same products, staged and pulled the same way.



01 · GREENFIELD

HEAT-RECOVERY BOILER

Behind every gas turbine: three-pressure HRSG with SCR and a duct burner. Feed pumps at 13.8 kV, drum-level and thermocouple runs, burner management in a Class I Div 2 skid.

707
CABLES PER HRSG TRAIN

157,000
FT OF CABLE PER TRAIN



02 · GREENFIELD OR RETROFIT

GAS-FIRED PACKAGE BOILER

The plant's auxiliary boiler, the same package sold to industry: burner and BMS, gas train with double block and bleed, FD fan on a VFD, feedwater pumps, its own MCC e-house.

83
CABLES PER PACKAGE BOILER

42,000
FT OF CABLE



03 · BROWNFIELD

MODULE REPLACEMENT OUTAGE

An HP evaporator module lifted out through the opened roof. The cable work rides the same outage window, so it has to be staged, pre-terminated and pulled fast.

Outage
CRITICAL PATH

SIMpull
+ PRE-TERMINATED HARNESSES

CABLE BY BOILER TYPE

	MV-105 15 KV FEED PUMPS	TC-ER 600 V POWER & VFD	ARMOR-X MC-HL GAS TRAINS (C1D2)	CONTROL TC-ER	INSTRUMENTATION PLTC · TC EXT.	SIMPULL · REEL ROLLERS	PRE-TERMINATED HARNESSES
HRSG + DUCT BURNER	●	●	●	●	●	●	●
PACKAGE BOILER		●	●	●	●	●	●
REPLACEMENT OUTAGE	○	●	●	●	●	●	●

● in scope ○ when the outage includes feed-pump or MV work

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE

HOW AN HRSG WORKS

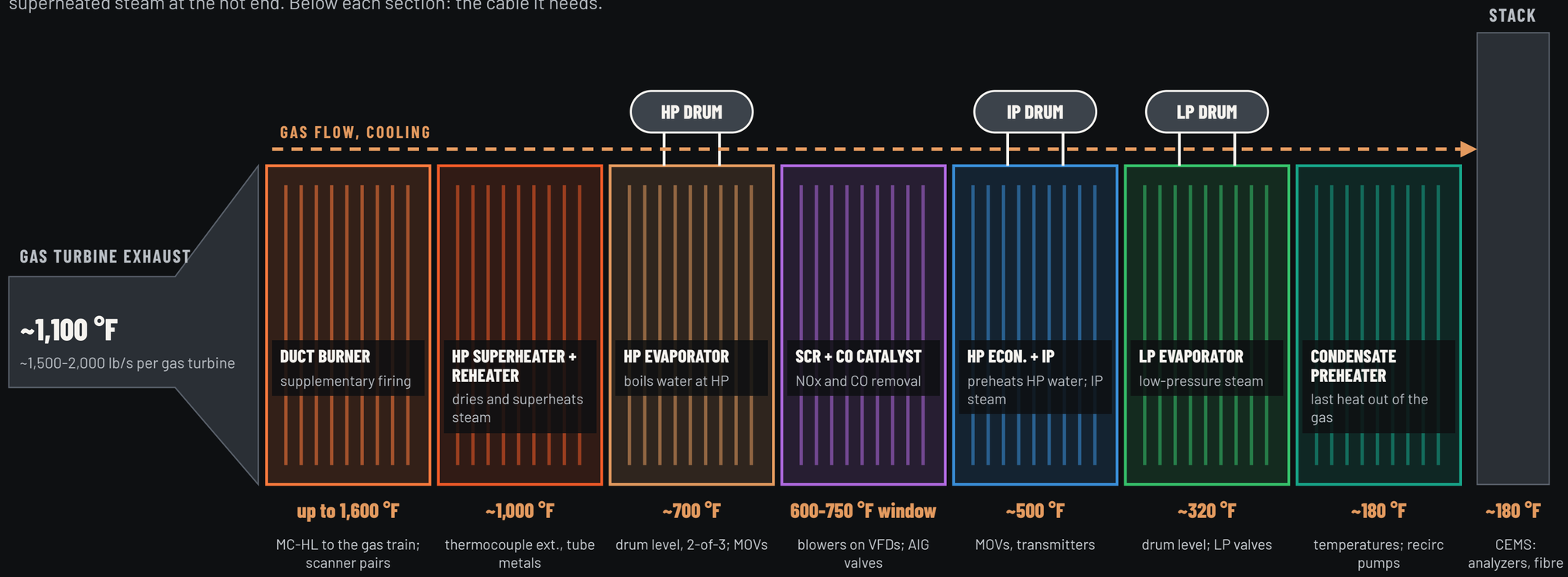
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.



Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

~1/3
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

~60 %+
combined-cycle efficiency, against ~40 % for the gas turbine alone

3 x 3
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

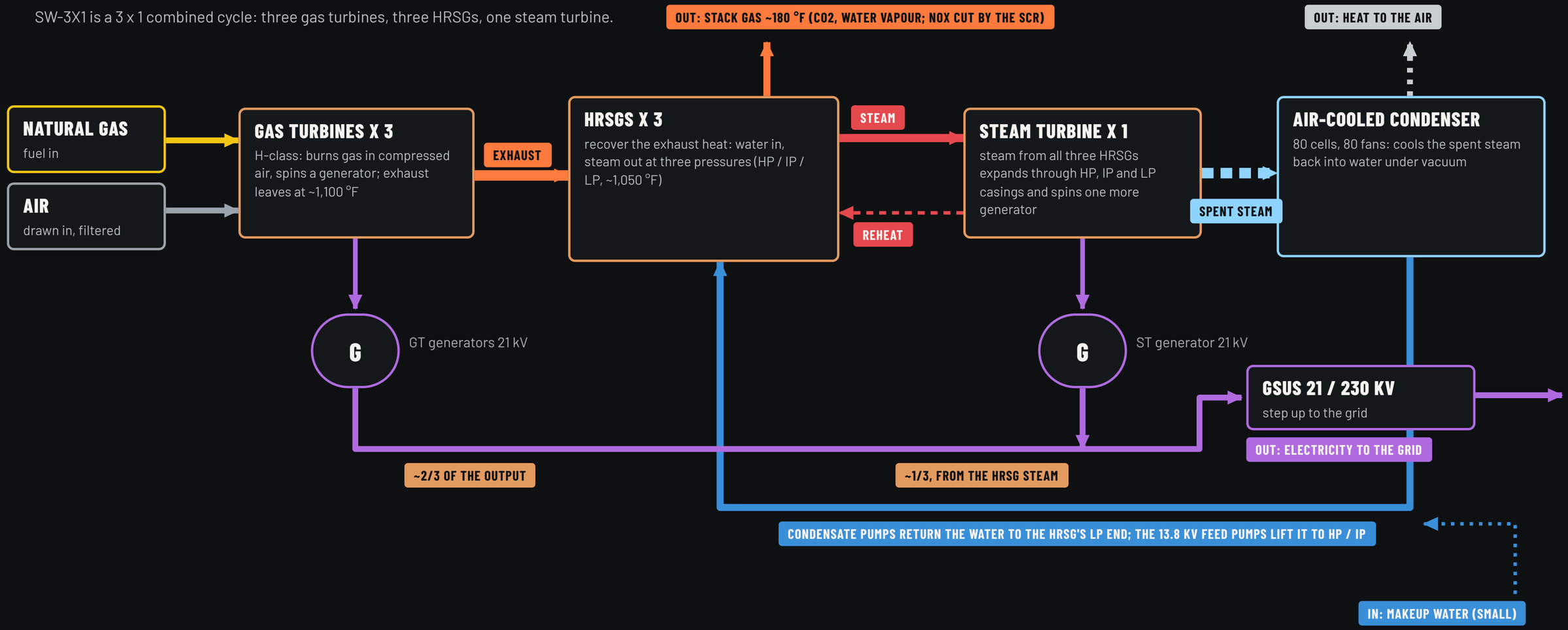
~1 hour
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.

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WHAT GOES IN, WHAT COMES OUT

SW-3X1 is a 3 x 1 combined cycle: three gas turbines, three HRSGs, one steam turbine.



IS IT STILL STEAM?

Leaving the turbine it is spent steam: ~100-150 °F, below atmospheric pressure, part water already. The air-cooled condenser turns it back into water.

A CLOSED LOOP

The same water goes round and round: pump, HRSG, turbine, condenser. Only a little makeup water replaces blowdown losses.

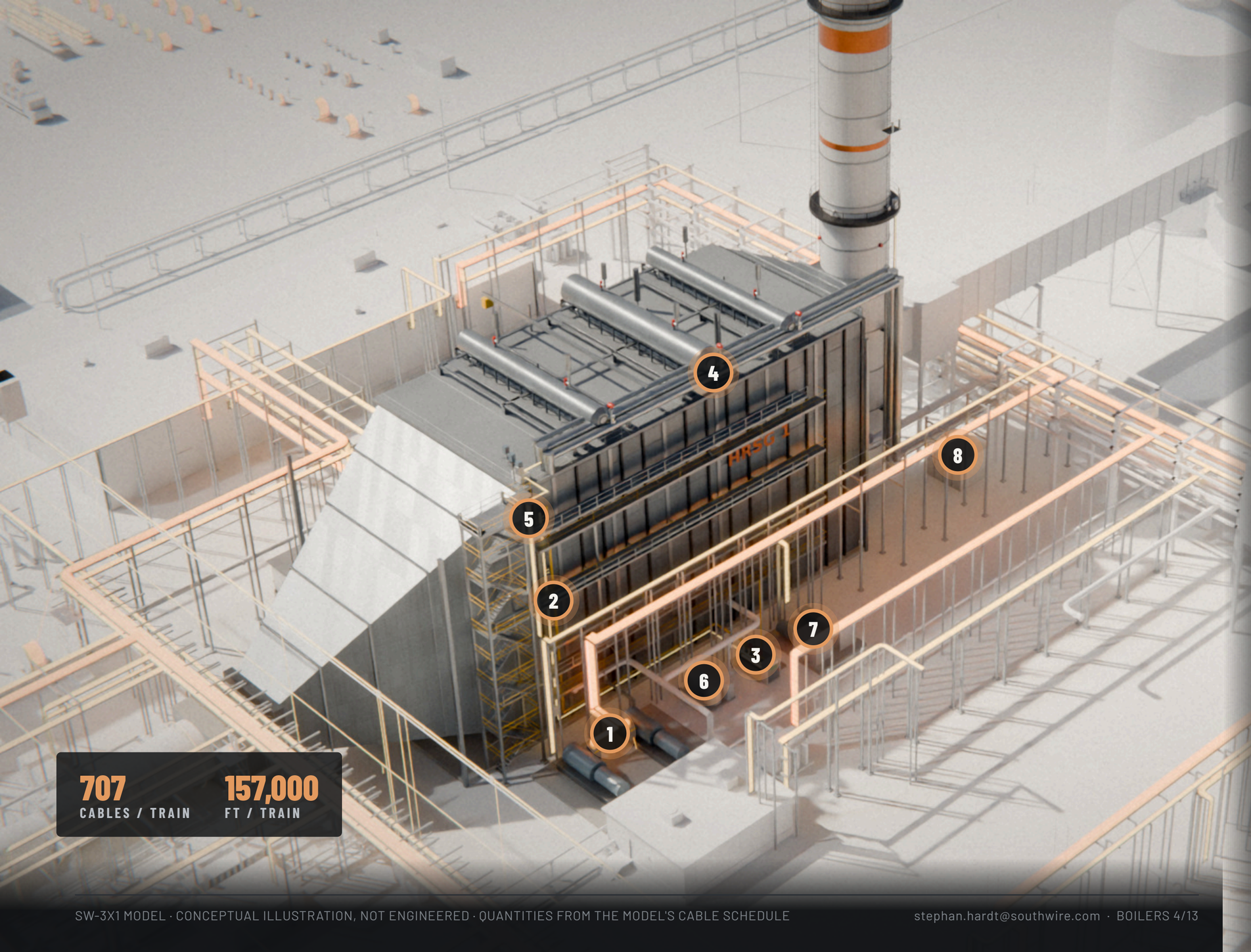
WHERE THE POWER COMES FROM

Fuel is burned only in the gas turbines (and duct burners). The steam turbine adds about half again on top, from heat that would otherwise go up the stack.

THE CABLE ON THE LOOP

21 kV isolated-phase bus to the GSUs; MV-105 to the feed pumps; VFD cable to 80 condenser fans; instruments on every pump, valve and level.

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707 CABLES / TRAIN

157,000 FT / TRAIN

BOILER SOLUTIONS · HRSG

THE HRSG, EVERY CABLE

One HRSG train and everything wired to it: cable in copper (MV) and amber (LV, control, instruments), the rest of the plant faded.

1. **FEED PUMPS, 13.8 KV**
   MV terminal boxes off the drop tray
   MV-105 15 kV [1]
2. **DUCT BURNER**
   gas header, 8 runners, igniters and flame scanners
   Control TC-ER [5] Instrumentation PLTC [6]
3. **DUCT-BURNER SKID AND BMS**
   double block and bleed, Class I Div 2
   ARMOR-X MC-HL [3] Control TC-ER [5]
4. **DRUMS AND ROOF VALVES**
   drum-level transmitters, motor-operated stop valves
   TC-ER power [2] Instrumentation PLTC [6]
5. **RISER TRAY TO THE ROOF**
   control, instrument and thermocouple cable
   Control TC-ER [5] Instrumentation PLTC [6] Thermocouple ext. [7]
6. **CHEMICAL FEED**
   metering-pump motors
   TC-ER power [2]
7. **SCR DILUTION-AIR BLOWERS**
   480 V motors
   TC-ER power [2] MC-HL VFD [4]
8. **CEMS SHELTER**
   analyzers and stack probes
   Instrumentation PLTC [6]

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THE TRAIN IN ELEVATION

The same train at a low angle from the west: e-house, the tray into the marshalling box, the ammonia grid, drums and stack. Cable in copper and amber, the rest faded.



1. **HRSG E-HOUSE R2A**
   MCCs and DCS I/O for the unit
   TC-ER power [2] Control TC-ER [5]
2. **EL 42 CONTROL TRAY**
   into the west-wall marshalling box
   Control TC-ER [5] Instrumentation PLTC [6]
3. **AMMONIA INJECTION GRID**
   AIG header and lances, SCR
   Control TC-ER [5] Instrumentation PLTC [6]
4. **BLOWDOWN TANK**
   level and valve control
   Control TC-ER [5] Instrumentation PLTC [6]
5. **DRUMS AND ROOF VALVES**
   level transmitters, MOVs
   TC-ER power [2] Instrumentation PLTC [6]
6. **STACK AND CEMS PORTS**
   sample lines, probes
   Instrumentation PLTC [6]
7. **GT EXHAUST TO THE HRSG**
   inlet duct thermocouples
   Thermocouple ext. [7]

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INSIDE THE ROOM

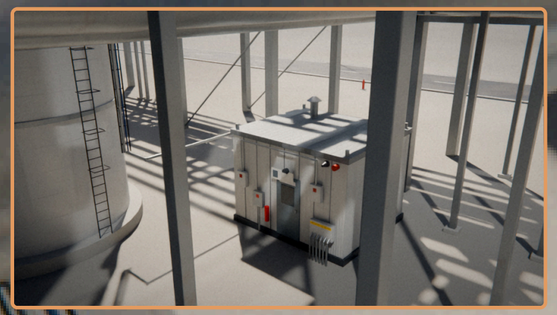
Roof and two walls lifted off the HRSG 1 e-house: every cable leg, transformer to section.

1. **13.8 KV PRIMARY**
   MV-105 off the EL 36 tray into T-R2A · 3 cables, 660 ft
   MV-105 15 kV [1]
2. **480 V SECONDARY**
   10 sets of 4 x 1/C 500 kcmil · 40 cables, 4,800 ft
   XHHW-2 1/C 500 kcmil
3. **MAIN BREAKER**
   the 40 cables drop into LC-R2A
   XHHW-2 1/C 500 kcmil
4. **LC-R2A FEEDERS**
   to MCCs, VFDs, UPS and panels
   TC-ER power [2]
5. **MCC DROPS**
   power and control into each bucket
   TC-ER power [2] **Control TC-ER** [5]
6. **VFD AND MCC-2**
   SCR blowers, chemical feed, blowdown
   TC-ER power [2] **MC-HL VFD** [4]
7. **MARSHALLING**
   home-run multipairs from the field JBs
   Instrumentation PLTC [6] **Thermocouple ext.** [7]
8. **OUT TO THE FIELD**
   power via the north wall, instruments via the roof
   TC-ER power [2] **Control TC-ER** [5] **Instrumentation PLTC** [6]
9. **UPS, DC AND PANELS**
   125 V DC, 120 V AC UPS, lighting
   TC-ER power [2] **Control TC-ER** [5]



Inside: LC-R2A and MCC-R2A-1 (right), VFDs (left), trays overhead

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Outside: conduit stub-ups, XP boxes, strobe and horn, HVAC



Up the stack: probes and junction box at EL 103



Stack top: obstruction lights, lightning protection

BOILER SOLUTIONS · EMISSIONS

THE STACK AND THE CEMS SHELTER

The analyzer shelter at each stack proves the plant's emissions. Roof and two walls lifted: every cable to it.

1. **CABLE ENTRY**
   conduit stub-ups into an XP junction box
   TC-ER power [2] Control TC-ER [5] Instrumentation PLTC [6]
2. **OVERHEAD LADDER TRAY**
   power, control, pairs, fire alarm, fibre
   TC-ER power [2] Control TC-ER [5] Instrumentation PLTC [6]
3. **ANALYZER RACKS**
   NOx / O2, CO / CO2, NH3 slip
   TC-ER power [2] Instrumentation PLTC [6]
4. **DATA ACQUISITION PLC**
   UPS-backed, fibre to the control room
   Instrumentation PLTC [6] Fibre
5. **SAMPLE CONDITIONING**
   chiller, pumps, filters, line controller
   TC-ER power [2]
6. **HEATED SAMPLE LINE**
   from the stack probes, through the bulkhead
   TC-ER power [2] Instrumentation PLTC [6]
7. **TRANSFORMER AND PANEL**
   480 / 208-120 V for HVAC, line, analyzers
   TC-ER power [2] SIMpull THHN
8. **FIRE ALARM, STROBE, HORN**
   loop back to the F&G panel in R2A
   FPLR fire alarm

31 CABLES / SHELTER

7,200 FT / SHELTER

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SWITCHGEAR ROOM TO THE HRSG

Every leg, from the 13.8 kV breaker to the field devices, by cable application: Southwire product (numbers as on the offering slide), cable count and footage from the model's schedule.

GSU-1 · UAT-1
21 / 13.8 kV



SWITCHGEAR ROOM R1

SWGR-13.8-1

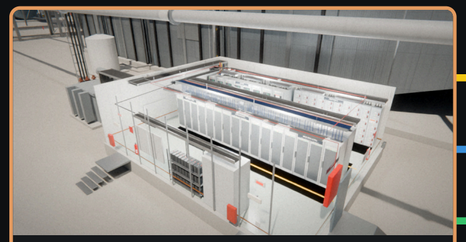
13.8 kV metal-clad: the feed-pump breakers and the T-R2A transformer feeder

MV-105 15 KV · 6 CABLES · 8,400 FT · 1

PRIMARY · MV-105 · 3 CABLES · 660 FT

T-R2A 13.8/ 0.48

SECONDARY · 1/C 500 KCMIL · 40 CABLES · 4,800 FT



HRSG E-HOUSE R2A

LC-R2A · MCCS · VFDS · DCS

480 V switchgear, starters, drives, UPS / DC, marshalling

In the room: lighting and receptacles, 2,280 ft THHN; panel feeder, control, fire-alarm loop, 330 ft

MAIN CONTROL ROOM
DCS servers, CEMS data

DCS FIBRE · 1 CABLE · 730 FT

FEED PUMPS BFP-1A / 1B, 13.8 KV
MV-105 15 kV · 6 cables · 8,400 ft 1

MOTORS, MOVS, HEATERS: SCR BLOWERS, CHEMICAL FEED, BLOWDOWN, CEMS
TC-ER 600 V power · VFD · 78 cables · 18,390 ft 2 · 4

VALVES, MOVS, BURNER MANAGEMENT, LOCAL STATIONS
Control TC-ER · ARMOR-X MC-HL · 97 cables · 21,730 ft 5 · 3

FIELD JUNCTION BOXES TO TRANSMITTERS, FLAME SCANNERS, RTDS
Instrumentation PLTC / ITC · RTD · 439 cables · 86,600 ft 6

GAS-PATH, TUBE-METAL AND DUCT THERMOCOUPLES
Thermocouple extension · 84 cables · 19,680 ft 7

T-R2A FANS, CONTROLS, TEMPERATURE / GAS MONITORING
TC-ER power · control · PLTC · 38 cables · 3,600 ft 2 · 5 · 6

CEMS SHELTER: DATA TO THE CONTROL ROOM, DETECTION, LIGHTING
Fibre · fire alarm · THHN · 3 cables · 2,580 ft —

THE HRSG 1 TRAIN: 707 CABLES, ~157,000 FT

+ FEEDING THE ROOM: 86 CABLES, 12,400 FT



R2A also serves the GT1 unit loads, so the T-R2A primary and secondary, the transformer auxiliaries, the DCS fibre and the room's own wiring are shown here but are not in the train total. Schedule rows: A-MV-0015, A-P-0425 and the rest by tag.

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED · QUANTITIES FROM THE MODEL'S CABLE SCHEDULE

LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

LEVEL	EQUIPMENT · RATING	FEEDS · CURRENT	CABLE IT DRIVES
230 kV	**GSU-1, 21 / 230 kV step-up**	Plant to the grid, overhead line	**Overhead conductor (outside the HRSG scope)**
21 kV	**GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA**	UAT-1 secondary 1,674 A into SWGR-13.8-1	**4 sets of 3 x 1/C 750 kcmil MV-105**
13.8 kV	**SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant**	Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A	**3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A**
480 V	**T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs**	Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles	**10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV**
120 V AC / 125 V DC	**UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels**	DCS, F&G node, breaker trip and close, burner-management power	**1/C #6 and control TC-ER 14 AWG**
Signals	**4-20 mA / HART, thermocouple millivolts, 24 V DC discretes**	Drum level, pressures, flame scanners, gas-path and tube-metal temperatures	**439 instrument + 84 thermocouple-extension cables, shielded pairs**

750 kW
connected at 480 V on one HRSG train, 78 circuits

2 x 5 MW
feed pumps: the largest motors on the train, at 13.8 kV

3,007 A
on the T-R2A secondary: 40 cables of 500 kcmil

74 %
of the train's cables are instrument or thermocouple pairs (68 % of the footage)

8.4 / 24.6
cal/cm^2 arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.

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WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

APPLICATION · WHERE	WHAT THE SPECIFICATION CALLS FOR	SOUTHWIRE
MV POWER, 15 KV FEED-PUMP MOTORS; THE T-R2A PRIMARY	1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS	**MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302**
480 V FEEDERS AND POWER T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS	Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct	**XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252**
TRAY CABLE: POWER AND CONTROL EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG	Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test	**Control TC-ER 14 AWG multiconductor · SPEC 45053**
HAZARDOUS AREA: GAS TRAIN, DUCT BURNER DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES	Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings	**ARMOR-X MC-HL, welded armor · SPEC 45352**
VARIABLE-FREQUENCY DRIVES SCR DILUTION-AIR BLOWERS, FD FANS	VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks	**ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352**
INSTRUMENTATION, 4-20 MA / HART DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS	**Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray	**Instrumentation PLTC, SPOS · SPEC 43303**
THERMOCOUPLE EXTENSION GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES	Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing	**Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350**
FIRE ALARM, FIBRE, BUILDING WIRE F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING	FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83)	**SIMpull THHN / THWN-2 · fire-alarm and fibre cable**

CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.

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PAIN POINTS AND THE ANSWER

TECHNICAL PAIN POINTS

- **MOSTLY SIGNAL CABLE** ~74 % of an HRSG's cables are instrument or thermocouple pairs; bad shield grounding or routing next to drives and MV makes noisy drum-level and flame-scanner signals that can trip the unit
- **THERMOCOUPLE EXTENSION ERRORS** ~84 runs per train: wrong alloy, reversed polarity or copper substituted; found as a bad tube-metal reading after startup
- **HEAT AND SUN** Hot casings, roof penthouses, drum platforms: 90-105 °C ratings, sunlight-resistant jackets, derating
- **MOVEMENT AND CYCLING** The HRSG grows inches when hot and cycles daily: slack at module joints, vibration, termination fatigue
- **GAS IN THE BOILER** Duct burners and fuel skids make Class I Div 2 areas; NFPA 85 burner management; conversions add it
- **BIG PARALLEL FEEDERS** 3,007 A transformer secondary = 40 cables of 500 kcmil; equal lengths in congested tray
- **DRIVES** SCR blowers and fans on VFDs: bearing currents and voltage spikes without VFD-rated cable
- **SCOPE GAPS** Vendor junction boxes, EPC home runs: legs fall between the two (our own schedule missed the 480 V secondaries)
- **SCHEDULE AND LABOUR** Cable is on the critical path, above all in an outage; electricians are scarce; no room to stage reels

VALUE PROPOSITION

- **ONE SUPPLIER, WHOLE SCOPE** MV-105, TC-ER, ARMOR-X MC-HL and VFD, instrumentation, thermocouple extension, THHN, fire alarm: one PO, one spec-compliance package (IEEE 1202, UL 1277 / 2225 / 13)
- **FASTER PULLS: SIMPULL** Pre-lubricated jacket cuts pulling force and time; parallel conductors on one reel go in one pull
- **STAGED FOR THE OUTAGE** Cut-to-length reels labelled per pull, SIMpull Cradles for one-person payoff: ready before the trip
- **ONE CABLE FOR HAZARDOUS AREAS** ARMOR-X MC-HL is rated Div 2 and Div 1: fewer conduits and seals, unaffected by reclassification
- **THE RIGHT CONSTRUCTION** VFD-rated cable for drives, individually shielded pairs, matched-alloy colour-coded thermocouple extension
- **ENGINEERING UP FRONT** Ampacity, voltage drop, spec review; takeoff ready: ~707 cables, ~157,000 ft per train, ~472,000 ft for three
- **SUSTAINABILITY** Reel returns and scrap recycling for cable cut back in brownfield outages

"On an HRSG, cable is three-quarters instrumentation and fully on the critical path. One spec-compliant supply, staged and fast to pull: wired on time, reading right the first time."

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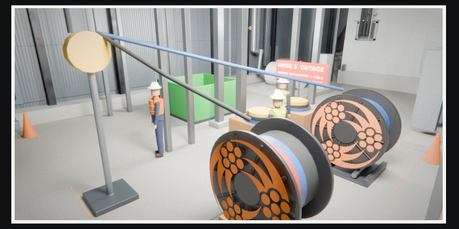
HRSG 3: an HP evaporator module lifted out through the opened roof by a crawler crane; the new module waits on a lowboy on the ring road.

BOILER SOLUTIONS · REPLACEMENT

THE OUTAGE CLOCK IS THE COST

On a boiler or HRSG replacement every hour of the outage costs. Cable rides the same window, so it moves off the critical path.

- **STAGE BEFORE THE TRIP** cut-to-length reels on SIMpull Cradles and reel rollers, labelled per pull, at the unit before the outage starts
- **PRE-TERMINATE** factory-made harnesses with QR tags land on the marshalling boxes: test, not terminate, on the critical path
- **PULL CLEAN** SIMpull reels and a sheave at the HRSG wall: one crew, no reel jacks, tension recorded
- **CUT BACK AND RECYCLE** old cable segregated into recycling bins as the module comes out
- **CONVERSIONS** new burner management and gas trains in Class I Div 2 run in ARMOR-X MC-HL



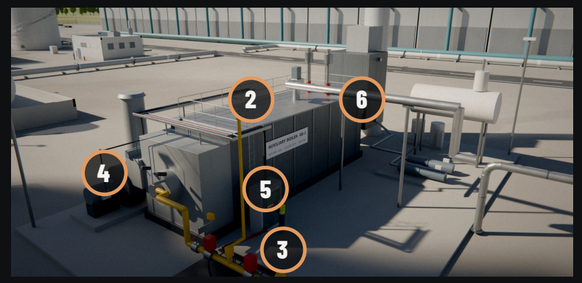
Cable work at the HRSG e-house



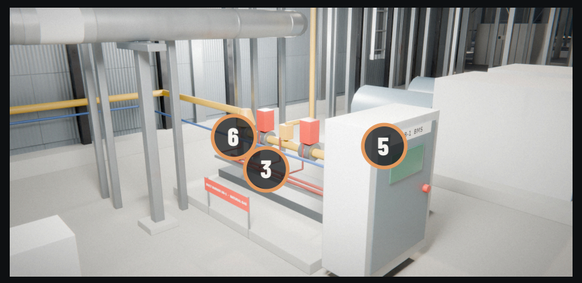
Old module on the hook

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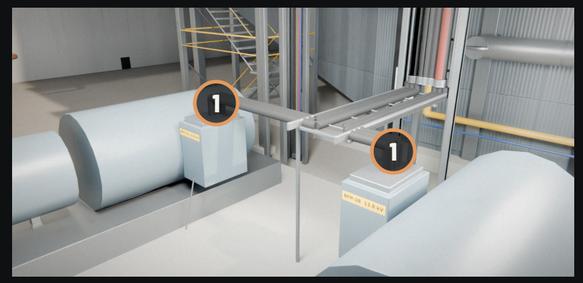
THE CABLE AT EVERY SPOT



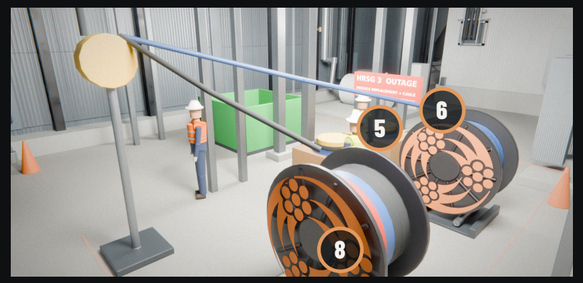
Package boiler AB-1



HRSG duct-burner skid



HRSG feed pumps, 13.8 kV



Outage cable work, HRSG 3

	PRODUCT	WHERE IT RUNS	SPEC	SOUTHWIRE STOCK # · SIZE	SPEC SHEET
1	**MV-105 15 KV** 1/C Cu, 220 mil NL-EPR 133 %, tape shield, SIMpull PVC	Feed-pump motors, 13.8 kV	46302	95604599 · 500 kcmil	southwire.com ›
2	**TYPE TC-ER POWER** 3/C Cu XHHW-2 + ground, PVC, 600 V	Fans, pumps, MCC feeders	45252	95305999 · 12 AWG 3/C	southwire.com ›
3	**ARMOR-X MC-HL** Cu XLPE XHHW-2, welded armor, PVC, 600 V, Class I Div 2	Gas trains, burner skids	45352	89052799 · 8 AWG 4/C	southwire.com ›
4	**ARMOR-X MC-HL VFD** 3/C Cu XLPE XHHW-2 + ground, VFD rated	FD fan on its drive	45352	89051899 · 3/0 3/C	southwire.com ›
5	**CONTROL TC-ER** Cu XLPE XHHW-2, 600 V	BMS panels, valves, MOVs	45053	89067701 · 14 AWG 5/C	southwire.com ›
6	**INSTRUMENTATION PLTC / ITC** 300 V, shielded pairs + overall shield (SPOS)	Flame scanners, drum level, transmitters	43303	per pair count	southwire.com ›
7	**THERMOCOUPLE EXTENSION** PVC, PLTC multi-pair (KX on request), 105 °C	HRSG gas-path and tube temperatures	42345	per construction	cabletechsupport.southwire.com ›
8	**SIMPULL CRADLE** for 51 / 61 in SIMpull Reels, one-person payoff	Outage staging and pulls	SPR-CRDL	67606001	southwire.com ›

Stock numbers are one catalogue size per product, from Southwire's product listings; every size and construction is on the same SPEC. Confirm current stock with Southwire customer service before quoting.

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